SLIDE 01 • SYSTEM REQUIREMENTS

Off-Grid Solar System for a Rural Health Centre

Dongola / Argo Island, Northern State, Sudan (19.16° N, 30.48° E)

Nicholas Khor
ID: 20617556

DAILY DEMAND

21.6 kWh/day

7,884 kWh annual energy requirement ($E = P \times t$)

PEAK DEMAND

2.0 kW

Occurs between 08:00 – 12:00 daytime operating block

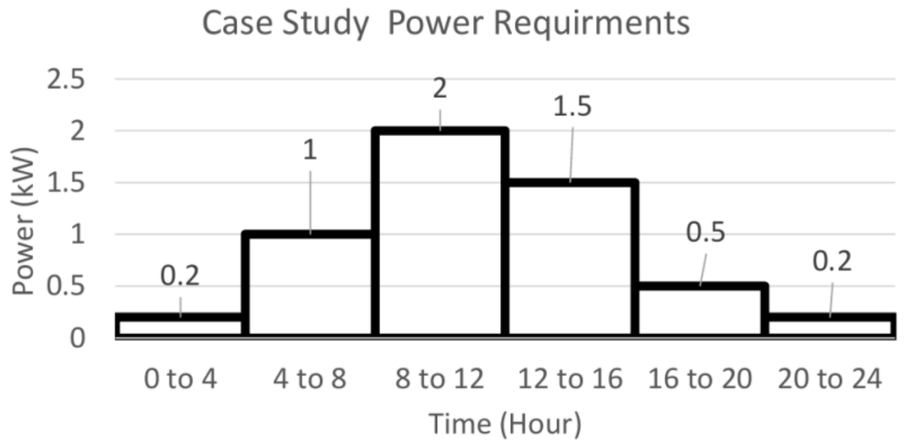


Figure 1: Power required at each four-hour interval.

Figure 1: Coursework power profile across 4-hour intervals over 24 hours

✓ **65% Daytime Solar Alignment (14.0 kWh)**

Majority of clinic operations (08:00 – 16:00) align directly with peak daytime solar irradiance.

🌙 **35% Night-time Battery Storage (7.6 kWh)**

Essential overnight loads (16:00 – 08:00) for vaccine refrigeration and medical lighting.

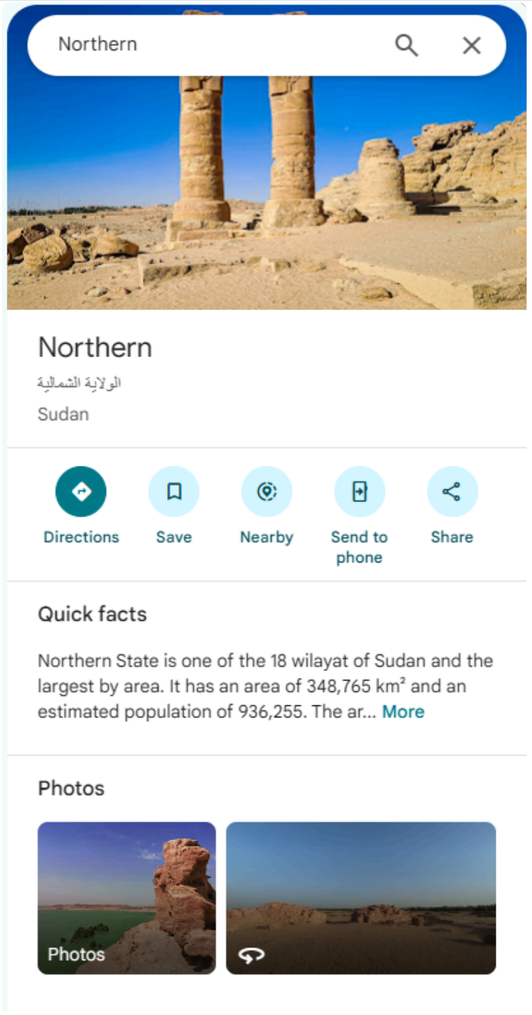
SLIDE 02 • GEOGRAPHY & CLIMATE

Site Selection & Local Microclimate

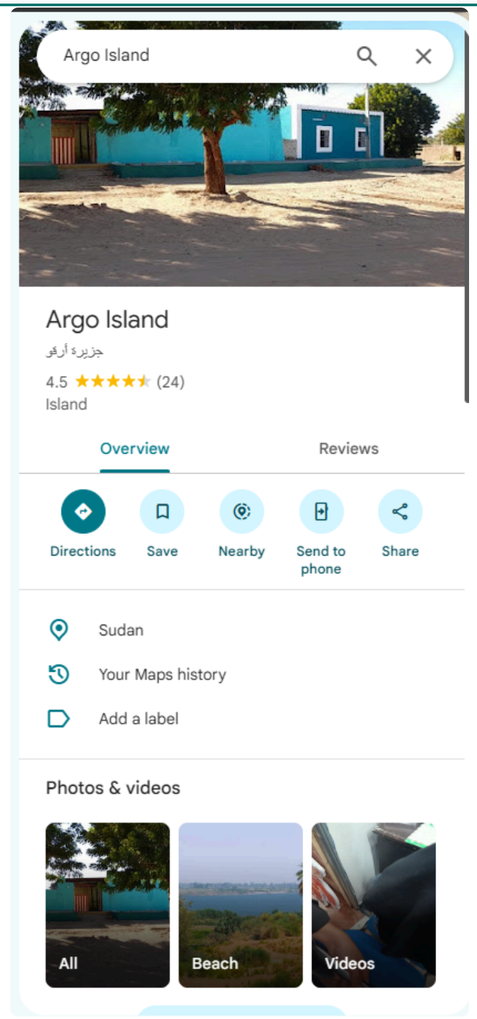
Applying the 1° Shift: Uninhabited Desert Dunes → Real Riverine Community

Northern State, Sudan

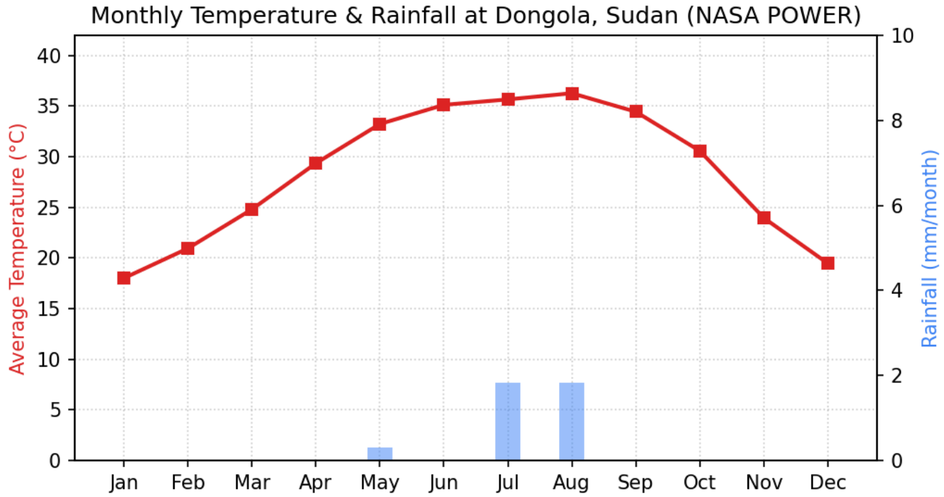
19.16° N, 30.48° E



Original Site (18.7°N, 29.6°E)
Empty Sahara sand dunes, 85 km west of Nile



Selected Village (19.16°N, 30.48°E)
Argo Island community, health clinic & road access



NASA POWER: Monthly 2m Air Temperature & Precipitation at Dongola

AVERAGE TEMPERATURE

28.8°C

High ambient heat (T_{amb}) requires PV derating

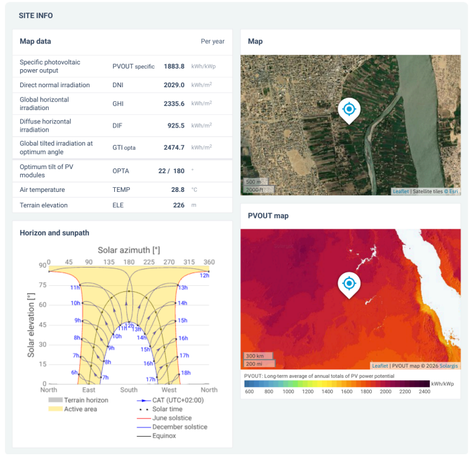
ANNUAL RAINFALL

< 4 mm/yr

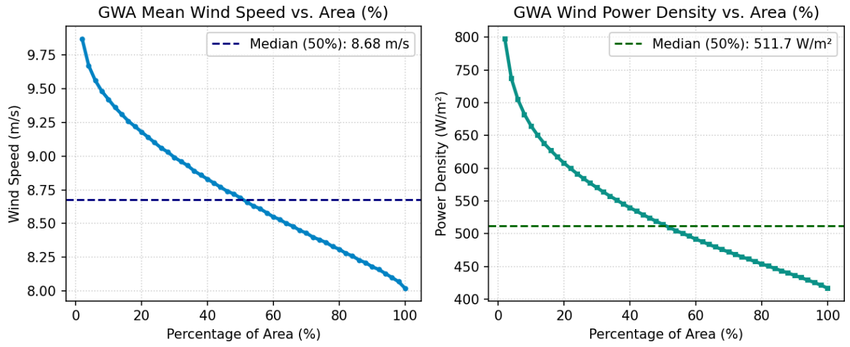
Hyper-arid desert environment; virtually zero cloud cover

Renewable Energy Potential Screening

High-Level Assessment Across All 5 Technologies Covered in Module



Global Solar Atlas: GTI = 2,474.7 kWh/m²/yr (6.78 PSH/day)



Global Wind Atlas: 100m Speed = 8.69 m/s (514 W/m²)

Resource	Measured Evidence at Dongola	Assessment
1. Solar PV	**6.78 PSH/day (GTI 2,475 kWh/m²/yr at 22° tilt); 1,884 kWh/kWp specific yield**	**1st Choice (Selected)**
2. Wind Power	NASA 10m wind: **5.11 m/s**; GWA 100m wind: **8.69 m/s** (514 W/m²)	**2nd Choice (Comparison)**
3. Hydro Power	Annual rainfall < 4 mm/yr; Nile plain has flat elevation and zero hydraulic head	**Unsuitable**
4. Geothermal	No tectonic rift or volcanic thermal anomalies within 600 km	**Unsuitable**
5. Wave & Tidal	Landlocked inland desert (>650 km inland from Red Sea)	**Impossible**

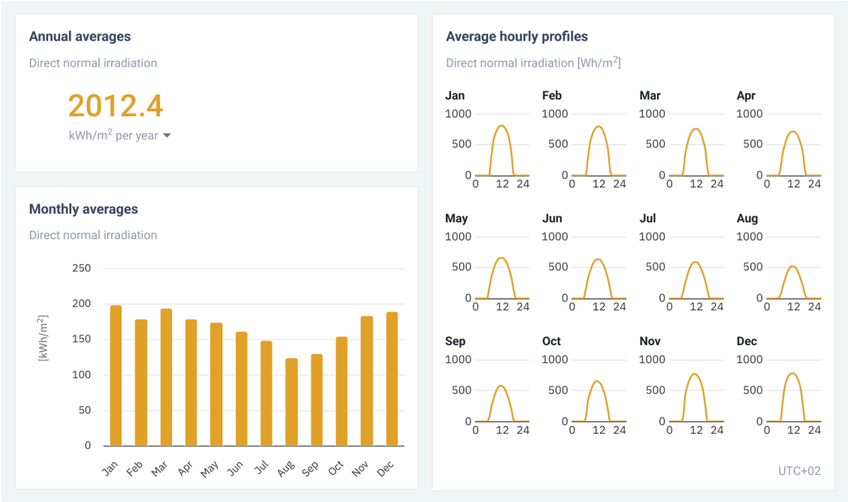
SLIDE 04 • OPTION EVALUATION

Head-to-Head Evaluation

Trade-Off Review

Solar PV (1st Choice) vs. Wind Power (2nd Choice)

Qualitative & Quantitative Justification for Primary Clinic Supply



Global Solar Atlas: Predictable daily solar bell curves all 12 months

Criterion	Solar PV (Preferred)	Small Wind (2nd Choice)
1. Resource Reliability	Constant daily sun: **6.78 PSH/day** at 22° tilt; zero cloudy months	NASA 10m wind: 5.1 m/s; cubic power law ($P \propto v^3$) intermittency
2. Load Synchronisation	Directly powers **65% of clinic load** during daylight hours (08:00 – 16:00)	Unpredictable daytime lulls require significantly larger battery capacity
3. Desert Durability	**Zero moving parts**; only periodic dry dust wiping required	Moving blades, bearings, and gears suffer heavy Saharan sand abrasion
4. Installed Economics	Low modular cost (~$0.30/Wp); simple ground-mounted aluminium racks	High capital cost: 10–15 m guyed mast, concrete base & crane logistics

SLIDE 05 • SYSTEM ARCHITECTURE

System Block Diagram & Energy Flow

Coupled DC-Bus Architecture with Li-Ion Storage & Single-Phase Inverter

230V AC • 50 Hz

Single-Phase Off-Grid

STAGE 1: GENERATION

PV Array (21 Panels)

- 21 × SANYO 210Wp modules
- 3 parallel strings × 7 series
- Tilted 22° South (GTI = 6.78 PSH)

`4,410 Wp STC Output`

STAGE 2: DC BUS SPLIT

Charge Controller

- Cable & MPPT efficiency: 95%
- Daytime: routes 14.0 kWh to clinic
- Diverts 7.6 kWh excess to battery

`21.6 kWh/day Delivered`

STAGE 3: ENERGY STORAGE

Li-Ion BESS (16 kWh)

- 8 × 2 kWh modules (48V bank)
- 95% round-trip efficiency
- 2× Safety buffer for healthcare

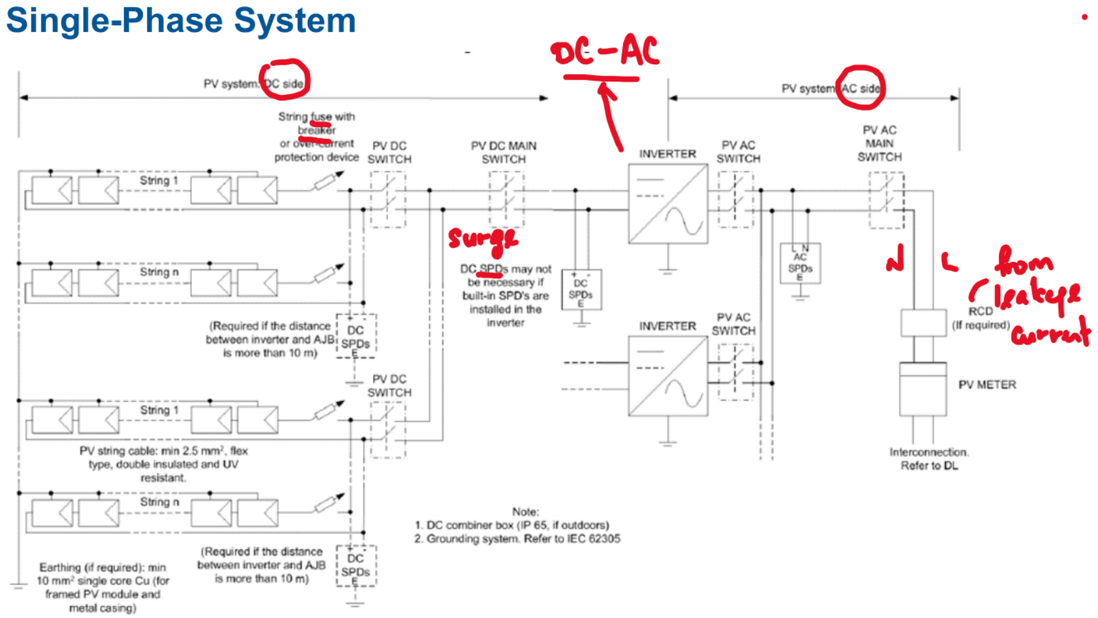
`7.6 kWh Nightly Drain`

STAGE 4: POWER INVERSION

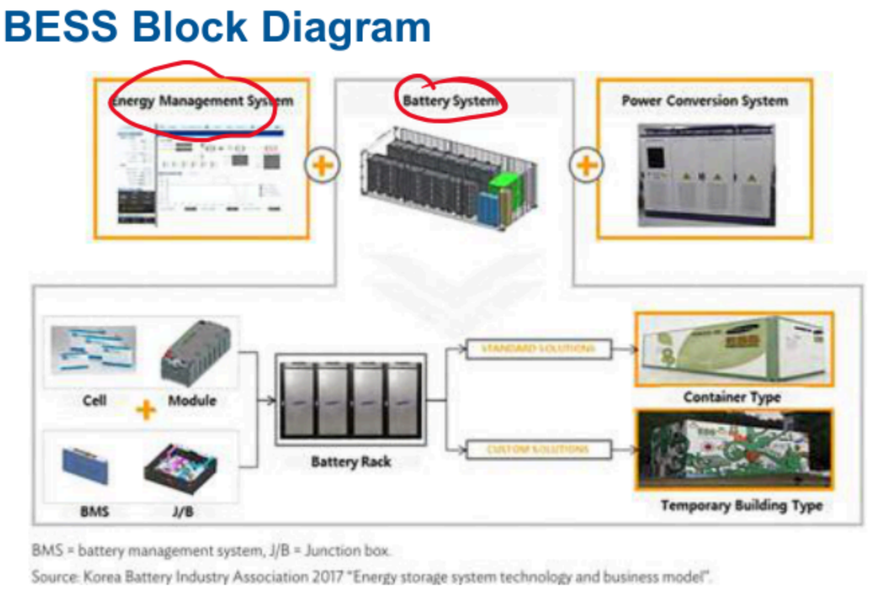
Fronius IG 40 Inverter

- Max efficiency: 94.3%
- Rated AC: 3,500 W (Peak load = 2.0 kW)
- Single-phase 230V AC, 50 Hz

`Health Centre Loads`



Lecture 3: Single-phase PV circuit & protective isolation layout



Lecture 5: BESS charge controller & DC bus architecture

SLIDE 06 • MATHEMATICAL SIZING

PV Array Sizing & Inverter Matching

Energy-Constraint Sizing Using Lecture 3 Equations & Hardware Datasheets

4.41 kWp PV Array

Fronius IG 40 Inverter

CALCULATED ARRAY STC

4,225 Wp

Minimum capacity required (P_{arr_stc})

INSTALLED ARRAY POWER

4,410 Wp

21 × SANYO HIP-210NKHE1 (210 Wp)

1 Annual Required Energy: $E_{req} = 7{,}884\ \text{kWh/yr}$

Calculated as 21.6 kWh/day × 365 days for 100% off-grid reliability.

2 Temperature Derate: $k_{tem} = 0.9136$

Derived from $T_{amb} = 28.8°C$ and SANYO coefficient $\beta = -0.30\%/°C$.

3 Inverter Matching: Fronius IG 40

Max DC input (4,410 W) matches array exactly; rated 3,500 W AC exceeds 2.0 kW peak load.

	Models HIP-xxxNKHE1		
Electrical data	**215**	**210**	**205**
Maximum power (Pmax) [W]	215	210	205
Max. power voltage (Vpm) [W]	42.0	41.3	40.7
Max. power current (Ipm) [A]	5.13	5.09	5.05
Open circuit voltage (Voc) [V]	51.6	50.9	50.3
Short circuit current (Isc) [A]	5.61	5.57	5.54
Warranted min. power (Pmin) [W]	204.3	199.5	194.8
Maximum over current rating [A]	15		
Output power tolerance [%]	+10/-5		
Max. system voltage [Vdc]	1000		
Temperature coeff. of Pmax [%/˚C]	-0.30		
Temperature coeff. of Voc [V/˚C]	-0.129	-0.127	-0.126
Temperature coeff. of Isc [mA/˚C]	1.68	1.67	1.66

Note 1: Standard test conditions: Air mass 1.5, Irradiance = 1000 W/m², Cell temperature = 25˚C.
Note 2: The values in the above table are nominal.

SANYO HIP-210NKHE1 Datasheet (210 Wp, 41.3V Vmp, −0.30%/°C)

TECHNICAL DATA: FRONIUS IG

INPUT DATA	IG 15	IG 20	IG 30	IG 40	IG 50[1)]	IG 60 HV
DC maximum power at cos φ = 1	1,610 W	2,150 W	2,850 W	4,410 W	4,950 W	5,380 W
Max. input current ($I_{dc\ max}$)	10.8 A	14.3 A	19.0 A	29.4 A	33.0 A	35.8 A
Min. input voltage ($U_{dc\ min}$)	150 V					
Feed-in start voltage ($U_{dc\ start}$)	170 V					
Nominal input voltage ($U_{dc,r}$)	280 V					
Max. input voltage ($U_{dc\ max}$)	500 V				530 V	
MPP voltage range ($U_{mpp\ min}$ – $U_{mpp\ max}$)	150 V – 400 V					
Number of DC inputs	5					

OUTPUT DATA	IG 15	IG 20	IG 30	IG 40	IG 50[1)]	IG 60 HV
AC nominal output ($P_{ac,r}$)	1,300 W	1,800 W	2,500 W	3,500 W	4,600 W	4,600 W
Max. output power	1,500 W	2,000 W	2,650 W	4,100 W	4,600 W	5,000 W
Max. output current ($I_{ac\ max}$)	6.5 A	8.7 A	11.5 A	17.8 A	20.0 A	21.7 A
Grid connection ($U_{ac,r}$)	1-NPE 230 V					
Min. output voltage ($U_{ac\ min}$)	180 V					
Max. output voltage ($U_{ac\ max}$)	270 V					
Frequency (f_r)	50 Hz / 60 Hz					
Frequency range (f_{min} – f_{max})	47 Hz – 65 Hz					
Distortion factor	< 3 %					
Power factor (cos $\varphi_{ac,r}$)	1					

Fronius IG 40 Datasheet (4,410 W DC max, 3,500 W AC, 94.3% eff)

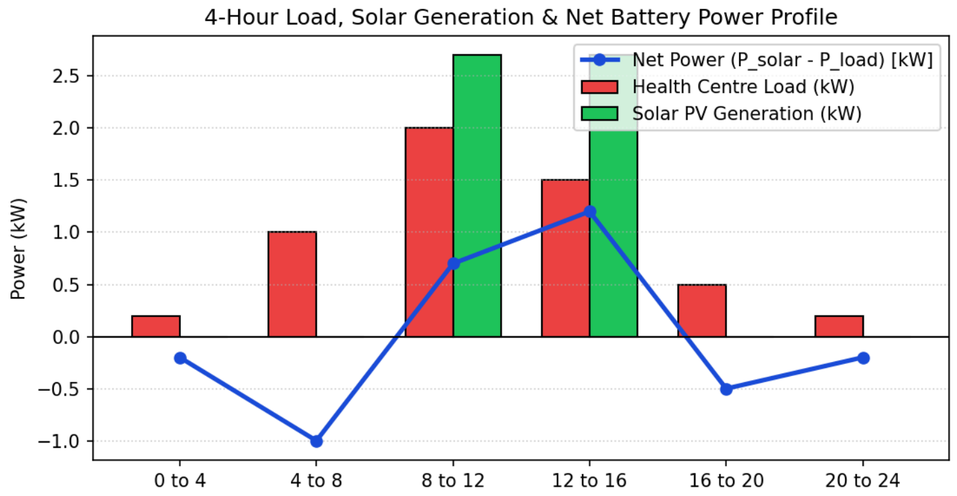
SLIDE 07 • STORAGE DESIGN

Battery Energy Storage System (BESS)

4-Hour Store vs. Drain Energy Balance & Technology Selection

16 kWh Li-Ion Bank

95% Round-Trip Eff.



4-Hour Load Demand vs. Solar Generation ($P_{net} = P_{solar} - P_{load}$)

Time Block	Load Energy	Solar Energy	Store (Excess)	Drain (Night)
00:00 – 08:00	4.8 kWh	0 kWh	—	**4.8 kWh**
08:00 – 16:00	**14.0 kWh**	**21.6 kWh**	**7.6 kWh**	**—**
16:00 – 24:00	2.8 kWh	0 kWh	—	**2.8 kWh**
Total 24 Hours	**21.6 kWh**	**21.6 kWh**	**7.6 kWh Stored**	**7.6 kWh Drained**

NIGHTLY DRAIN

7.6 kWh

Normal overnight requirement (16:00 – 08:00)

SELECTED CAPACITY

16 kWh

8 × 2 kWh Li-Ion units (2× healthcare safety buffer)

SLIDE 08 • ECONOMIC EVALUATION

System Cost Breakdown & Payback Period

Verified CAPEX, OPEX & Payback Against Rural Diesel Baseline ($0.40/kWh)

Initial CAPEX: $8,500

Payback: 2.88 Years

Component	Quantity	Verified Cost	Lifetime
Solar Panels (SANYO 210W)	21 panels (4.41 kWp)	**$1,500**	20–25 yrs (IRENA 2024)
Li-Ion BESS (2 kWh units)	8 units (16 kWh)	**$4,000**	10–15 yrs (NREL 2025)
Fronius IG 40 + Controller	1 unit (3.5 kW AC)	**$1,200**	10–12 yrs
Mounting, BOS & Sudan Delivery	Cables, frame, logistics	**$1,800**	20 yrs (UNDP Sudan)
Total Initial CAPEX	**Turnkey System**	**$8,500**	**Complete Installation**
Year 11 Replacement	1× Battery + Inverter	**$5,200**	Refurbished for yrs 11–20
Annual Maintenance (OPEX)	Dust wiping & inspection	**$200/yr**	$4,000 over 20 years

ANNUAL NET SAVING

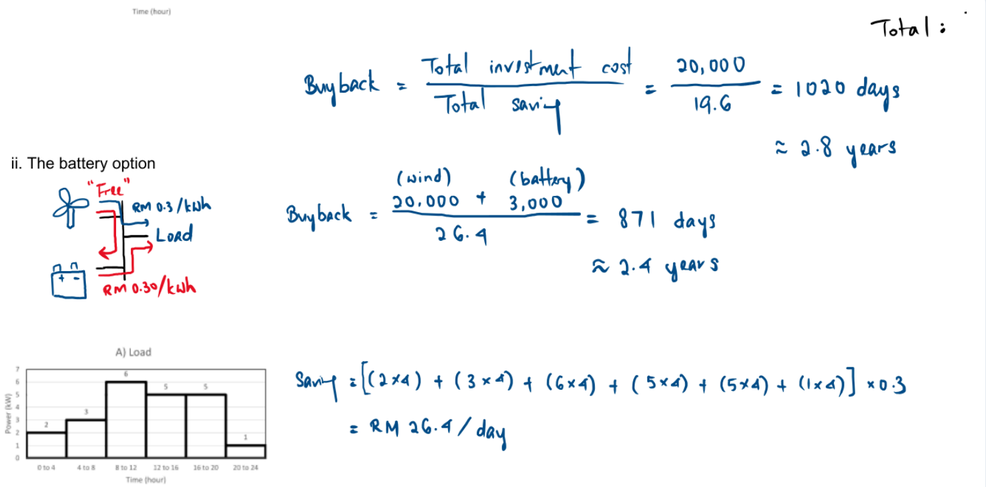
$2,954

Displacing $0.40/kWh diesel ($3,154 – $200 OPEX)

PAYBACK PERIOD

2.88 Years

~1,050 operating days to full financial ROI



Lecture 5 Reference: Payback = Initial CAPEX / Annual Net Savings = $8,500 / $2,954 = **2.88 years (~1,050 days)**

SLIDE 09 • ENVIRONMENTAL & TECHNOLOGY COMPARISON

CO_2 Savings & 5 System Comparisons

Proposed Solar PV + BESS vs. 3.5 kW Baseline Diesel Generator

141.9 Tonnes CO_2 Saved

94.7% Emissions Cut

20-YEAR CO_2 SAVED

141.9 Tonnes

Solar Lifecycle: 7.9 t vs. Diesel: 149.8 t

Zero Operating Emissions

Generates 157,680 kWh over 20 years with zero fuel burned.

55,188 Litres Diesel Saved

Eliminates fragile fuel supply routes in rural Sudan.

#	Criterion	Solar PV + Battery	Diesel Generator
1	**Initial CAPEX**	$8,500 (Panels, battery & inverter)	**Diesel: $1,800 (79% cheaper)**
2	**Annual Running Cost**	**Solar: $200/yr (94% cheaper)**	$3,150/year (2,760 L fuel + oil changes)
3	**20-Year Total Cost**	**Solar: $17,700 ($0.11/kWh)**	**$68,400 ($0.43/kWh) – Saves $50,700!**
4	**20-Year CO_2 Footprint**	**Solar: 7.9 tonnes (94.7% cut)**	149.8 tonnes (From burning 55,188 L diesel)
5	**Noise & Supply Security**	**Solar: 0 dB Silent & Free Sun**	75–85 dB continuous noise; fuel shortage risks

SLIDE 10 • ACADEMIC RIGOUR

IRENA • NREL • UNDP

Audited & Corrected

Critical AI Evaluation & Key References

Fact-Checking Generative AI Outputs Against Peer-Reviewed Benchmarks

Dimension	Generative AI Output	Verified Ground-Truth Fix
What AI Got Right	Solar panel pricing (~$0.30/Wp); zero operating emissions	**Validated** Matches IRENA 2024 global report
What AI Got Wrong	Quoted $150/kWh EV battery; assumed 1 battery lasts 20 yrs	Corrected to **$250/kWh** rack battery + **$5,200** replacement at Yr 11
Hidden Assumptions	Assumed free local shipping; assumed 100% generator efficiency	Added **$1,800** remote Sudan BOS/logistics; used realistic 0.35 L/kWh fuel burn
20-Yr Cost Correction	**$7,420 (Unrealistically low)**	**$17,700 (Realistic turnkey lifecycle cost)**

1 **Meteorological & Solar Atlases**

World Bank / Solargis *Global Solar Atlas* & NASA POWER Agroclimatology (Dongola 19.16°N, 30.48°E).

2 **Equipment & Hardware Benchmarks**

IRENA (2024) *Renewable Power Generation Costs*; NREL / NLR (2025) *Battery Storage Cost Projections*.

3 **Sudan Logistics & Lifecycle Carbon**

UNDP Sudan (2024) *Solar Energy Value Chains in Sudan*; NREL LCA PV Harmonization Study.